

DATA ANALYST ASSESSMENT

Retail Pricing & Product-Mix Analysis

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Final Assessment Report

This report consolidates the completed assessment answers, data-quality treatment, business findings, recommendations, dashboard evidence, management presentation outline, and AI-use disclosure from the submitted workbook.

1. Executive Summary

The analysis uses the Diamonds dataset to identify observable price drivers, product-segment differences, value concentration and data-quality risks relevant to retail pricing and assortment decisions.

KPI / Finding	Result
Raw dataset	53,940 rows × 10 columns
Analysis-ready dataset	53,775 rows × 15 columns
Carat-price correlation	0.922
Premium vs Ideal average price	Premium 4,579 vs Ideal 3,462 (+32.3%)
2+ ct segment	3.5% of records / 13.2% of value proxy
Top 10% by price	34.4% of total price value

Core management takeaway: carat is the strongest observable price driver, while high-value products are concentrated in a relatively small part of the assortment. Cut and clarity comparisons should therefore be interpreted with carat/product mix in mind.

2. Dataset & Business Problem (Q1–Q2)

Item	Answer
1. Dataset name	Diamonds — ggplot2 built-in dataset
2. Source and URL	ggplot2 / Tidy verse documentation — https://ggplot2.tidyverse.org/reference/
3. Number of rows and columns	53,940 rows × 10 columns
4. Brief description	53,940 observations of round-cut diamonds with carat, cut, color, clarity, depth, table, price and physical dimensions (x, y, z).
5. Why selected	Exceeds the 50,000-record requirement and supports pricing, segmentation, mix and data-quality analysis.
6. Business opportunities/problems	Identify price drivers, high-value segments, assortment concentration and data-quality risks for pricing and merchandising decisions.

Item	Definition
A. Business problem / opportunity	How can a digital retail business use product characteristics and quality segments to improve pricing and product-mix decisions while protecting data quality?
B1	Which product attributes are most strongly associated with price?
B2	How does average price differ across cut, color and clarity segments?
B3	Which carat bands contribute the largest share of the revenue proxy?
B4	Are high-priced products concentrated in a small part of the assortment?
B5	Which records or segments require caution because of unusual data values?
C1. Hypothesis	Larger carat size is the strongest observable driver of price.
C2. Hypothesis	Upper-end products contribute disproportionately to the revenue proxy compared with their record share.
C3. Hypothesis	Quality labels do not explain price independently of carat/product mix.

3. Data Cleaning & Preparation (Q3)

Change	What changed	Why necessary	What if not done
Remove duplicate rows	Removed 146 exact duplicate records.	Prevents repeated observations from inflating counts and value totals.	Counts, segment shares and value totals could be overstated.
Flag/remove zero physical dimensions	Found 20 records with at least one of X/Y/Z equal to zero; removed them from analysis-ready data.	Zero dimensions make calculated volume invalid.	Size/volume analysis could contain impossible observations.
Create calculated fields	Added Volume_mm3 and Price_per_Carat.	Enables size normalization and value-efficiency comparisons.	Analysis would be less useful for pricing decisions.
Create business categories	Created Size_Category and Price_Tier.	Makes results easier for management to interpret.	Continuous measures are harder to translate into actions.
Retain legitimate high-price values	Did not automatically delete high prices as outliers.	Premium products can legitimately have extreme values.	Removing them could understate premium economics.

4. Key Findings & Business Implications (Q4)

Insight	Evidence	Business Impact	Recommendation
1. Carat is the strongest price driver	Carat-price correlation = 0.922; volume-price correlation = 0.904.	Supports size-based pricing and assortment planning.	Use carat bands as a primary pricing lens.
2. Premium cut has higher average price than Ideal	Premium avg = 4,579; Ideal avg = 3,462; difference = 32.3%.	Potentially higher ticket sizes in Premium.	Review premium-cut pricing while controlling for carat.
3. 2+ ct is a small but valuable segment	2+ ct = 3.5% of records and 13.2% of value proxy.	High-value inventory has outsized commercial importance.	Protect availability and pricing discipline for high-carat products.
4. Top 10% by price are disproportionately valuable	Top 10% contribute 34.4% of total price value.	Small premium group deserves focused attention.	Create a premium monitoring view.
5. Ideal is common but not highest priced	Ideal is 40.0% of processed records, but avg price is 3,462.	Avoid optimizing only for assortment volume.	Manage using both volume and value contribution.

5. Unexpected Findings & Analytical Risks (Q5–Q6)

Question	Answer
1. What did you initially expect?	Higher clarity/cut quality would map fairly directly to higher prices.
2. What did the data actually show?	Average price is not monotonic with clarity; SI2 has the highest average price in this dataset. Ideal is the most common cut but not the highest average-price cut.
3. Why do you think this happened?	Carat and product mix strongly influence price, so segment averages can be confounded by size.
4. What additional analysis did you perform?	Compared average price and price-per-carat across segments and checked correlations with carat, volume, depth and table.
5. What conclusion can reasonably be drawn?	Quality labels should not be interpreted independently of carat/product mix; causal pricing claims require a controlled model.

Issue / Risk	Evidence	Potential Effect	How Handled / Accounted For
Exact duplicate records	146 exact duplicates.	Can inflate counts and value totals.	Removed exact duplicates.
Zero physical dimensions	20 records with at least one zero in X/Y/Z.	Invalid volume and suspicious measurements.	Removed from analysis-ready data.
Extreme values	Some observations have high price/carat/dimensions.	Automatic removal may delete legitimate premium products.	Retained unless violating a clear validity rule.
Scope limitation	No customer/channel/promotion n/cost/margin variables.	Cannot assess profitability or customer behavior.	Kept conclusions limited to product-level price and mix.
Confounding risk	Carat strongly correlates with price and varies across segments.	Segment price gaps may reflect mix, not causality.	Avoided causal claims; recommend carat-controlled analysis.

Two limitations: no margin/customer/channel/promotion data; descriptive analysis cannot establish causal pricing effects.

Conclusion that cannot safely be made: changing cut or clarity alone would cause the observed price difference, because carat/product mix is a major confounder.

6. Actionable Recommendations (Q7)

Priority	What should be done?	Supporting insight	Who should act?	Potential outcome	Success measures
1 — High	Use carat bands as a core pricing and assortment lens.	Carat is the strongest price correlate; 2+ ct has disproportionate value.	Pricing + Merchandising	Better focus on high-value inventory.	Average price, price per carat, high-carat availability, premium mix.
2 — High	Create a premium-product monitoring view for the top 10% by price.	Top 10% contribute ~34.4% of total price value.	Commercial / Category Management	More focused attention on highest-value products.	Top-decile value share, realized price, availability, markdown rate.
3 — Medium	Evaluate cut/clarity premiums only after controlling for carat.	Segment averages are mix-sensitive.	Pricing Analytics	More defensible pricing decisions.	Carat-adjusted premium, price-per-carat, model residuals.

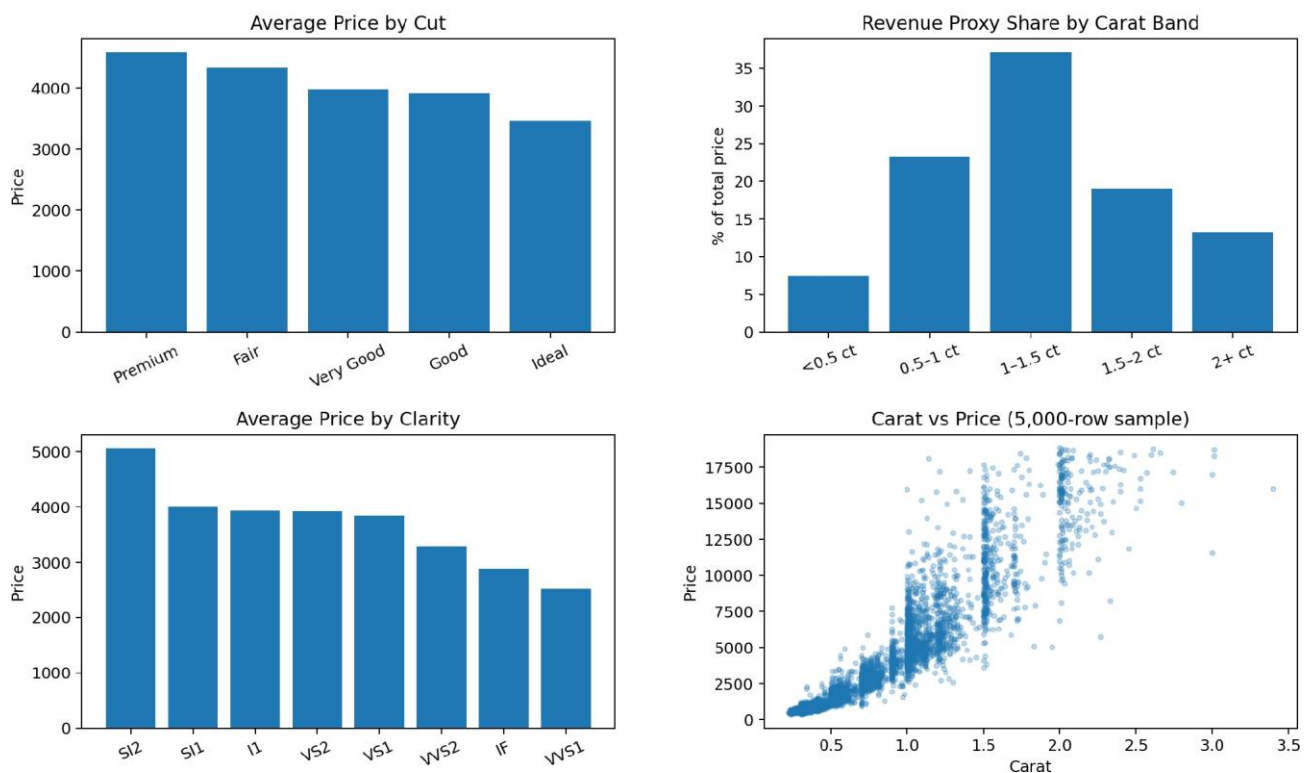
7. Interactive Dashboard (Q8)

Purpose: compact management view of price drivers, product mix and premium-value concentration.

Dashboard component	Purpose / detail
KPI 1	Processed records: 53,775
KPI 2	Average price: 3,931
KPI 3	Median price: 2,401
KPI 4	Carat-price correlation: 0.922
Visualization 1	Average price by Cut
Visualization 2	Revenue proxy share by Carat Band
Visualization 3	Average price by Clarity
Visualization 4	Carat vs Price scatter
Filters	Cut, Color, Clarity, Size_Category and Price_Tier

Dashboard screenshot from the submitted workbook:

Management Dashboard — Diamond Retail Pricing & Mix



Looker Studio report: <https://datastudio.google.com/reporting/f9897d87-2dbb-492c-bb08-bda61e9a2b31>

8. Management Presentation (Q9)

Slide	Content
Slide 1 — Business Problem	Improve pricing and assortment decisions using product-level characteristics.
Slide 2 — Data & Methodology	53,940 rows × 10 columns; cleaned to 53,775 analysis-ready rows; added calculated fields.
Slide 3 — Key Findings	Carat-price $r=0.922$; Premium avg price +32.3% vs Ideal; 2+ ct = 3.5% records / 13.2% value.
Slide 4 — Deep Dive	Top 10% by price contribute ~34.4% of total price value; larger carat bands have disproportionate value.
Slide 5 — Recommendations	Carat-aware pricing; premium monitoring; carat-controlled cut/clarity evaluation.
Slide 6 — Expected Business Impact	Focus on high-value inventory and improve pricing discipline; track realized price, price-per-carat and premium mix.
Slide 7 — Limitations & Next Steps	No margin/customer/channel data; next step is a carat-controlled pricing model with richer commercial data.

9. AI Usage Disclosure (Q10)

Item	Answer
1. AI tools used	ChatGPT / GPT-5.6
2. What used them for	Assessment structuring, analytical approach, Python logic, business-question drafting, insight wording and presentation structure.
3. One example where AI helped	Helped translate descriptive statistics into a management-focused narrative around carat-driven pricing and premium-value concentration.
4. One example where AI was verified/corrected	All key statistics were recalculated directly from the dataset in Python before being included.

AI was used as an assistive tool; key statistics were recalculated directly from the dataset before inclusion.

10. Supporting Segment Summaries

Cut Summary

Cut	Records	Avg_Price	Median_Price	Avg_Price_per_Carat
Premium	13737.0	4578.91	3180.0	4222.51
Fair	1597.0	4340.68	3282.0	3761.65
Very Good	12068.0	3980.92	2646.5	4014.44
Good	4888.0	3916.28	3038.0	3859.09
Ideal	21485.0	3462.15	1813.0	3922.38

Carat Band / Size Summary

Size_Category	Records	Revenue_Proxy	Avg_Price	Record_Share	Revenue_Share
<0.5 ct	18863.0	15837760.0	839.6204	0.3508	0.0749
0.5–1 ct	17473.0	49124876.0	2811.4735	0.3249	0.2324
1–1.5 ct	12028.0	78377045.0	6516.2159	0.2237	0.3707
1.5–2 ct	3544.0	40141238.0	11326.5344	0.0659	0.1899
2+ ct	1867.0	27920452.0	14954.7145	0.0347	0.1321

11. Color & Clarity Summaries

Color Summary

Color	Records	Avg_Price	Avg_Price_per_Carat
J	2802.0	5326.42	3827.23
I	5406.0	5079.84	3993.21
H	8266.0	4477.1	4006.83
G	11254.0	3999.09	4165.23
F	9517.0	3726.78	4136.11
D	6754.0	3172.59	3954.18
E	9776.0	3079.61	3806.94

Clarity Summary

Clarity	Records	Avg_Price	Avg_Price_per_Carat
SI2	9142.0	5054.53	4010.6
SI1	13030.0	3994.27	3849.27
I1	737.0	3927.3	2797.28
VS2	12225.0	3925.61	4081.47
VS1	8155.0	3841.3	4157.49
VVS2	5056.0	3286.53	4205.71
IF	1784.0	2870.57	4263.18
VVS1	3646.0	2522.99	3851.17

End of Final Assessment Report